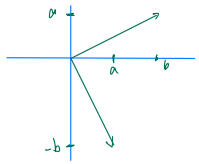
- Scale and Rotate

$$\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$$



rotate by $\arctan(\frac{b}{a})$

scale by $\sqrt{a^2+b^2}$

- Complex Eigenvalue & Eigenvectors

Suppose $Ax=\lambda x$ $A \in \mathbb{R}^{n\times n}$ $x \neq 0$ $\begin{cases} \lambda_1 = \delta + j\omega & x_1 = x_r + jx_c \\ \lambda_2 = \delta - j\omega & x_2 = x_r - jx_c \end{cases}$

$$Ax_1 = \lambda x_1 = (\delta + j\omega)(x_r + jx_c) = (\delta x_r - \omega x_c) + j(\omega x_r + \delta x_c)$$

$$\lambda x_2 = (\delta - j\omega)(x_r - jx_c) = (\delta x_r - \omega x_c) - j(\omega x_r + \delta x_c) = Ax_2$$

- Eigenvalue Decomposition (2x2 case)

Complex Eigenvalue	Real Eigenvalue
$\begin{cases} \lambda_1 = \delta + j\omega & x_1 = x_r + jx_c \\ \lambda_2 = \delta - j\omega & x_2 = x_r - jx_c \end{cases}$	$\begin{cases} \lambda_1 & x_1 \\ \lambda_2 & x_2 \end{cases}$

Decomposition

$$A = \begin{bmatrix} | & | \\ x_r & x_c \\ | & | \end{bmatrix} \begin{bmatrix} \delta & \omega \\ -\omega & \delta \end{bmatrix} \begin{bmatrix} | & | \\ x_r & x_c \\ | & | \end{bmatrix}^{-1}$$

$$A = \begin{bmatrix} | & | \\ x_1 & x_2 \\ | & | \end{bmatrix} \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} \begin{bmatrix} | & | \\ x_1 & x_2 \\ | & | \end{bmatrix}^{-1}$$

Why the decomposition make sense

$$A(x_r + jx_c) = (\delta x_r - \omega x_c) + j(\omega x_r + \delta x_c)$$

$$A\begin{bmatrix} | & | \\ x_r & x_c \\ | & | \end{bmatrix} = \begin{bmatrix} | & | \\ x_r & x_c \\ | & | \end{bmatrix} \begin{bmatrix} \delta & \omega \\ -\omega & \delta \end{bmatrix} \begin{bmatrix} | & | \\ x_r & x_c \\ | & | \end{bmatrix}^{-1} \begin{bmatrix} | & | \\ x_r & x_c \\ | & | \end{bmatrix}$$

$$\begin{bmatrix} | & | \\ Ax_r & Ax_c \\ | & | \end{bmatrix} = \begin{bmatrix} | & | \\ \delta x_r - \omega x_c & \omega x_r + \delta x_c \\ | & | \end{bmatrix}$$

real part of "eigen equation"

imaginary part of eigen equation

$$Ax_1 = \lambda_1 x_1 \quad Ax_2 = \lambda_2 x_2$$

$$A\begin{bmatrix} | & | \\ x_1 & x_2 \\ | & | \end{bmatrix} = \begin{bmatrix} | & | \\ x_1 & x_2 \\ | & | \end{bmatrix} \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} \begin{bmatrix} | & | \\ x_1 & x_2 \\ | & | \end{bmatrix}^{-1} \begin{bmatrix} | & | \\ x_1 & x_2 \\ | & | \end{bmatrix}$$

$$\begin{bmatrix} | & | \\ Ax_1 & Ax_2 \\ | & | \end{bmatrix} = \begin{bmatrix} | & | \\ \lambda_1 x_1 & \lambda_2 x_2 \\ | & | \end{bmatrix}$$

First eigen equation Second "eigen equation"

the decomposition decomposes A into 2 "eigen equations"

· Interpretation

$$Av = \begin{bmatrix} | & | \\ x_r & x_c \\ | & | \end{bmatrix} \begin{bmatrix} \alpha & w \\ -w & \alpha \end{bmatrix} \begin{bmatrix} | & | \\ x_r & x_c \\ | & | \end{bmatrix}^{-1} v$$

reassemble — rotate & scale — express v in (x_r, x_c) frame

A is rotation and scale in (x_r, x_c) coordinate

$$Av = \begin{bmatrix} | & | \\ x_1 & x_2 \\ | & | \end{bmatrix} \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} \begin{bmatrix} | & | \\ x_1 & x_2 \\ | & | \end{bmatrix}^{-1} v$$

reassemble — scale — express v in (x_1, x_2)

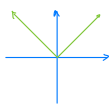
A is scaling in (x_1, x_2) coordinate

$\lambda = \alpha \pm jw \quad |\lambda| = \sqrt{\alpha^2 + w^2}$

$\begin{bmatrix} \alpha & w \\ -w & \alpha \end{bmatrix}$ → the scaling induced by A

Eg.

$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

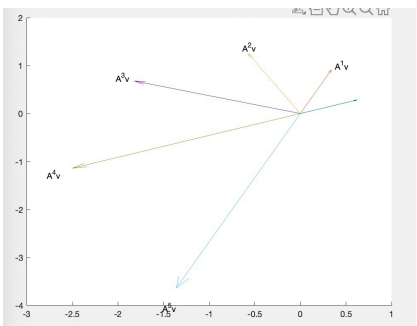


$\lambda_1 = 1+j \quad x_1 = \begin{bmatrix} 1 \\ -j \end{bmatrix} \quad x_r = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\lambda_2 = 1-j \quad x_2 = \begin{bmatrix} 1 \\ j \end{bmatrix} \quad x_c = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

$$A = \begin{bmatrix} | & | \\ x_r & x_c \\ | & | \end{bmatrix} \begin{bmatrix} \alpha & w \\ -w & \alpha \end{bmatrix} \begin{bmatrix} | & | \\ x_r & x_c \\ | & | \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$|\lambda| = \sqrt{2}$



· Real and Complex Eigenvalue (3×3 case)

Suppose $A \in \mathbb{R}^{3\times3}$, $Ax = \lambda x$, A has

1. a conjugate pair of complex eigenvalues $\begin{cases} \lambda_1 = \alpha + jw & x_1 = x_r + jx_c \\ \lambda_2 = \alpha - jw & x_2 = x_r - jx_c \end{cases}$
2. a real eigenvalue (λ_3, x_3)

$$A = \begin{bmatrix} | & | & | \\ x_r & x_c & x_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} \alpha & w & \\ -w & \alpha & \\ & & \lambda_3 \end{bmatrix} \begin{bmatrix} | & | & | \\ x_r & x_c & x_3 \\ | & | & | \end{bmatrix}^{-1}$$

$$A \begin{bmatrix} | & | & | \\ x_r & x_c & x_3 \\ | & | & | \end{bmatrix} = \begin{bmatrix} | & | & | \\ x_r & x_c & x_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} \alpha & w & \\ -w & \alpha & \\ & & \lambda_3 \end{bmatrix}$$

$$\begin{bmatrix} | & | & | \\ Ax_r & Ax_c & Ax_3 \\ | & | & | \end{bmatrix} = \begin{bmatrix} | & | & | \\ \alpha x_r - w x_c & w x_r + \alpha x_c & \lambda_3 x_3 \\ | & | & | \end{bmatrix}$$

Real Eigen Equation

Real part of complex "eigen equation"

Complex part of complex "eigen equation"

Suppose $v = a_r x_r + a_c x_c + a_3 x_3$

$$Av = \begin{bmatrix} | & | & | \\ x_r & x_c & x_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} \delta & \omega & \\ -\omega & \delta & \\ & & \lambda_3 \end{bmatrix} \begin{bmatrix} | & | & | \\ x_r & x_c & x_3 \\ | & | & | \end{bmatrix}^{-1} v$$

$$= \begin{bmatrix} | & | & | \\ x_r & x_c & x_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} \delta & \omega & \\ -\omega & \delta & \\ & & \lambda_3 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$= \begin{bmatrix} | & | & | \\ x_r & x_c & x_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} \begin{bmatrix} \delta & \omega \\ -\omega & \delta \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \\ \lambda_3 a_3 \end{bmatrix}$$

} the component on (x_r, x_c) plane is rotated & scaled

} component on x_3 axis is scaled

Eg.

$$A = \exp\left(0.3 \times \begin{bmatrix} -0 & -1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \right)$$

$\lambda_1 = 1.0534 + 0.3845j$ $\quad x_1 = \begin{bmatrix} -0.6894 \\ 0.1767 \\ 0.3690 \end{bmatrix} + \begin{bmatrix} 0 \\ -0.5338 \\ 0.2705 \end{bmatrix} j$

$\lambda_2 = 1.0534 - 0.3845j$ $\quad x_1 = \begin{bmatrix} -0.6894 \\ 0.1767 \\ 0.3690 \end{bmatrix} - \begin{bmatrix} 0 \\ -0.5338 \\ 0.2705 \end{bmatrix} j$

$\lambda_3 = 0.8195$ $\quad x_3 = \begin{bmatrix} 0.3445 \\ 0.5191 \\ 0.7811 \end{bmatrix}$

$|\lambda_1| = |\lambda_2| > 1$

the component on (x_r, x_c) plane grows

$|\lambda_3| < 1$

the component on x_3 axis decays

